

PLANT CITY, FL, USA

WMO: **720373**

Lat: **28.000N** Lon: **82.164W** Elev: **47** StdP: **100.76** Time Zone: **-5.00 (NAE)** Period: **06-19** WBAN: **92824**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	2.5	4.9	-6.3	2.2	10.9	-2.8	3.0	12.5	8.5	21.6	7.9	22.0	1.4	10	0.336	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	8	8.8	35.1	25.1	34.1	25.0	33.7	24.9	27.4	31.0	27.1	30.8	26.8	30.5	2.6	100	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	26.4	22.0	28.2	26.2	21.8	28.2	26.0	21.5	28.1	86.6	31.1	85.3	30.7	84.0	31.2	29.2	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	7.6	6.6	5.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(4)
(5)				WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	23.5	16.2	18.6	19.8	23.3	26.2	27.9	28.6	28.9	28.0	24.7	20.2	18.9	(6)
(7)			DBStd	5.43	4.91	4.83	3.91	2.63	2.01	1.36	1.18	1.11	1.21	3.19	4.02	4.64	(7)
(8)			HDD10.0	15	9	3	0	0	0	0	0	0	0	0	0	3	(8)
(9)			HDD18.3	271	98	53	33	2	0	0	0	0	0	4	27	53	(9)
(10)			CDD10.0	4933	202	243	304	400	503	538	578	587	539	454	307	278	(10)
(11)			CDD18.3	2146	33	59	78	152	245	288	319	328	289	200	84	69	(11)
(12)			CDH23.3	20632	219	439	666	1426	2520	2937	3342	3489	2854	1729	585	426	(12)
(13)			CDH26.7	8417	29	98	165	525	1166	1354	1489	1527	1216	631	134	82	(13)
(14)	Wind		WSAvg	2.2	2.5	2.4	2.7	2.6	2.5	2.0	1.7	1.7	1.9	2.2	2.2	2.3	(14)
(15)	Precipitation		PrecAvg	1370	69	65	82	67	74	222	208	218	192	66	45	55	(15)
(16)			PrecMax	1747	165	221	215	195	223	403	305	378	418	157	173	308	(16)
(17)			PrecMin	968	9	8	11	7	9	58	120	114	65	5	6	2	(17)
(18)			PrecStd	202	41	49	64	38	52	85	50	67	86	39	35	70	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		28.9	30.9	31.0	33.7	35.3	36.0	35.3	35.3	34.8	33.0	31.3	30.1	(19)
(20)			MCWB		21.0	21.3	21.4	22.6	22.5	24.7	25.5	26.0	25.2	24.1	24.2	23.0	(20)
(21)		2%	DB		27.5	29.0	29.9	32.3	34.0	34.8	34.9	34.9	33.8	32.4	29.8	28.7	(21)
(22)			MCWB		20.4	20.9	20.4	21.6	22.3	24.8	25.5	26.0	25.1	24.2	23.0	22.3	(22)
(23)		5%	DB		26.1	27.7	28.7	31.1	32.8	33.7	33.9	33.9	32.8	31.3	27.9	27.5	(23)
(24)			MCWB		19.5	20.2	20.2	21.3	22.5	24.8	25.4	26.0	24.8	23.9	21.4	21.5	(24)
(25)		10%	DB		24.0	26.3	27.4	29.9	32.3	32.7	32.8	32.8	32.3	30.1	27.1	26.1	(25)
(26)			MCWB		18.6	19.2	19.0	20.6	22.4	24.8	25.5	26.0	24.9	23.1	20.6	20.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		23.2	23.4	24.0	25.3	25.9	27.5	27.8	28.0	27.6	26.7	25.5	24.7	(27)
(28)			MCDB		26.4	26.2	27.7	29.3	29.6	31.0	31.2	31.0	30.7	29.3	29.2	27.6	(28)
(29)		2%	WB		22.0	22.5	22.8	24.1	25.2	27.0	27.3	27.5	27.1	25.9	24.3	23.8	(29)
(30)			MCDB		25.1	26.2	26.0	28.5	29.7	30.6	30.9	31.2	30.4	29.2	27.4	26.8	(30)
(31)		5%	WB		21.0	21.9	21.8	23.2	24.7	26.4	26.9	27.2	26.6	25.4	23.2	22.8	(31)
(32)			MCDB		23.8	25.8	25.5	27.7	29.2	30.3	30.7	31.0	29.8	29.0	26.2	25.6	(32)
(33)		10%	WB		20.0	21.0	20.8	22.5	24.1	26.0	26.5	26.8	26.1	24.8	22.2	22.1	(33)
(34)			MCDB		23.1	24.7	24.6	27.0	28.8	29.9	30.5	30.7	29.5	28.3	25.4	25.1	(34)
(35)	Mean Daily Temperature Range		MDBR		12.0	12.2	12.7	12.3	11.8	10.0	9.2	8.8	9.1	10.4	11.6	11.4	(35)
(36)		5% DB	MCDBR		12.3	12.0	12.9	13.2	12.6	11.1	10.1	9.6	9.6	9.9	11.1	11.0	(36)
(37)			MCWBR		6.5	5.1	4.7	4.3	3.8	3.5	3.2	3.1	3.2	3.5	5.2	5.5	(37)
(38)		5% WB	MCDBR		10.7	10.1	10.8	11.2	10.5	9.5	8.9	8.8	8.6	8.5	9.2	9.9	(38)
(39)			MCWBR		6.8	5.4	5.3	4.9	3.8	3.6	3.4	3.2	3.2	3.5	4.7	5.7	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.338	0.351	0.365	0.386	0.422	0.448	0.472	0.453	0.419	0.388	0.352	0.338	(40)
(41)		taud			2.519	2.494	2.443	2.353	2.307	2.312	2.246	2.316	2.388	2.449	2.496	2.546	(41)
(42)		Ebn at Noon			907	923	926	911	872	845	823	839	861	871	883	891	(42)
(43)		Edh at Noon			91	100	112	126	132	131	139	129	117	104	92	85	(43)
(44)	All-Sky Solar Radiation	RadAvg			3.53	4.23	5.33	6.35	6.67	5.98	5.75	5.54	5.05	4.66	3.81	3.25	(44)
(45)		RadStd			0.25	0.26	0.42	0.31	0.40	0.50	0.36	0.35	0.30	0.30	0.19	0.26	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (40 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	-4	-45	+158	+112	(47)

Nomenclature: See separate page